Nephrotic Syndrome 15 (MAGI2): A Comprehensive Disease Characterization

MONDO:0033262 · OMIM:617609 · Causal gene: *MAGI2* · Category: Mendelian (autosomal recessive podocytopathy)

Summary

Nephrotic Syndrome 15 (NS15) is an ultra-rare, autosomal recessive **podocytopathy** caused by biallelic loss-of-function variants in ***MAGI2*** *— membrane-associated guanylate kinase, WW and PDZ domain containing 2* (HGNC:18957; NCBI Gene 9863; UniProt Q86UL8; gene OMIM 606382; cytogenetic locus 7q21.11). The gene encodes an "inverted" MAGUK scaffold protein that organizes the podocyte **slit diaphragm**, the specialized cell–cell junction of the glomerular filtration barrier. When both *MAGI2* alleles are inactivated, podocytes lose a central scaffold required to anchor the slit-diaphragm backbone proteins nephrin and Neph1, to couple the junction to the actin cytoskeleton, and to sustain Rap1/RhoA small-GTPase signaling. The clinical result is **congenital-to-childhood-onset steroid-resistant nephrotic syndrome (SRNS)** with focal segmental glomerulosclerosis (FSGS) or diffuse mesangial sclerosis, progressing to end-stage kidney disease (ESKD).

The mechanistic basis of NS15 is unusually well supported for an ultra-rare disease, owing to convergent evidence from human patients, three independent mouse knockout models, zebrafish CRISPR knockouts, and cultured human podocytes. MAGI2 loss disrupts the **MAGI2–RapGEF2–Rap1** signaling complex and broader **RhoA/Rac1/Cdc42 small-GTPase network**, collapsing the podocyte actin cytoskeleton, and triggers **dendrin-** and **TGF-β/Smad3-driven podocyte apoptosis**. Human podocytes expressing patient variants show dramatic actin disorganization that is fully rescued by pharmacological Rap1 activation, pinpointing a druggable node. Podocyte-specific knockout mice develop massive albuminuria by 8 weeks and die by 20 weeks, faithfully recapitulating the human disease.

Because NS15 is monogenic, it carries two clinically important corollaries: patients respond poorly to immunosuppression (only ~4% complete response in a genetically stratified cohort) but characteristically **do not experience disease recurrence after kidney transplantation**,

making molecular diagnosis directly actionable for treatment and transplant planning. No disease-specific pharmacotherapy exists; management is supportive and renal-replacement-based, though AAV-mediated MAGI2 gene replacement has shown proof-of-concept protection in animal models.

Key Findings

Finding 1 — Biallelic MAGI2 loss-of-function causes NS15 (F001)

Recessive *MAGI2* mutations were established as a cause of congenital/steroid-resistant nephrotic syndrome in humans, mirrored by severe glomerulosclerosis in *MAGI2* knockout mice. MAGI2 is a MAGUK (membrane-associated guanylate kinase) scaffold protein expressed in podocytes, the specialized visceral epithelial cells of the kidney glomerulus. As stated directly in the primary literature: *"The essential role of membrane associated guanylate kinase 2 (MAGI2) in podocytes is indicated by the phenotypes of severe glomerulosclerosis of both MAGI2 knockout mice and in patients with congenital nephrotic syndrome (CNS) caused by mutations in MAGI2"* (PMID: 31171376). This dual human-and-mouse concordance anchors causality.

Finding 2 — Mechanism: MAGI2–RapGEF2–Rap1 signaling and actin cytoskeleton collapse (F002)

MAGI2 forms a complex with the guanine-nucleotide exchange factor **RapGEF2** to activate the small GTPase **Rap1**. CNS-associated MAGI2 variants lose RapGEF2 binding and fail to activate Rap1. Podocyte-specific RapGEF2 deletion phenocopies the MAGI2-knockout glomerulosclerosis, placing RapGEF2–Rap1 directly downstream of MAGI2. Critically, human podocytes expressing MAGI2 CNS variants *"show severe abnormalities of cellular morphology and dramatic loss of actin cytoskeletal organization, features completely rescued by pharmacological activation of Rap1 via a non-MAGI2 dependent upstream pathway"* (PMID: 31171376). The full rescue by Rap1 activation defines the causal pathway and nominates Rap1 as a therapeutic target.

Finding 3 — Mouse knockouts recapitulate the human disease (F003)

Podocyte-specific MAGI-2-knockout mice show slit-diaphragm disruption, foot-process abnormalities, and podocyte apoptosis/loss. The pathology follows a defined time course: *"these pathologic changes manifested as massive albuminuria by 8 weeks of age and*

glomerulosclerosis and significantly higher plasma creatinine levels at 12 weeks of age; all MAGI-2-KO mice died by 20 weeks of age" (PMID: 28539383). MAGI-2 loss decreased expression and promoted nuclear translocation of the pro-apoptotic protein **dendrin**. An independent constitutive knockout confirmed the mechanism: *"Magi-2 null mice presented with progressive proteinuria as early as 2 wk postnatally, which coincided with loss of nephrin expression in the glomeruli"* (PMID: 25271328). Two independent knockout strategies converging on proteinuria and nephrin loss strongly corroborate causality.

Finding 4 — Monogenic SRNS: poor immunosuppression response, no transplant recurrence (F004)

In the UK National Study of Nephrotic Syndrome cohort, patients with monogenic SRNS (the category that includes MAGI2) showed only **4% complete response** to intensified immunosuppression versus **25%** in genetic-testing-negative patients, and had **no post-transplant disease recurrence**: *"Patients with monogenic steroid-resistant nephrotic syndrome had a poor therapeutic response and no post-transplant recurrence"* (PMID: 32317330). Molecular diagnosis therefore directly predicts poor immunosuppression response and favorable transplant outcome — a key actionable implication.

Finding 5 — Variant spectrum: biallelic truncating variants abolish podocyte MAGI2 (F005)

The index report (Bierzynska et al., 2017, *JASN*) identified MAGI2 mutations by whole-exome sequencing in **3 patients from 2 families** with congenital/childhood-onset SRNS: *"We detected two unique frameshift mutations and one duplication in three patients (two families); two siblings shared the same homozygous frameshift mutation, whereas one individual with sporadic SRNS exhibited compound heterozygosity"* (PMID: 27932480). Two variants introduced premature stop codons and one caused read-through of the normal stop codon; immunohistochemistry showed absent or diminished podocyte MAGI2, confirming loss-of-function with autosomal recessive inheritance.

Finding 6 — MAGI2 acts in a RhoA/Cdc42 small-GTPase network; partial treatment sensitivity in some patients (F006)

Ashraf et al., 2018 (*Nat Commun*) showed that *"mutations in six different genes (MAGI2, TNS2, DLC1, CDK20, ITSN1, ITSN2) as causing NS in 17 families with partially treatment-sensitive NS (pTSNS). These proteins interact and we delineate their roles in Rho-like small GTPase (RLSG) activity"* (PMID: 29773874). Patient mutants are deficient in Rho-like GTPase activity;

knockdown of MAGI2 in cultured podocytes reduced migration rate, and dexamethasone modulated RhoA signaling in this module. This places MAGI2 within a broader RhoA-regulating network, explains partial treatment sensitivity in some families, and nominates therapeutic targets.

Finding 7 — MAGI2 localizes nephrin/Neph1; complete loss is neonatally lethal (F007)

Yamada et al., 2021 (*Kidney Int*) showed that *"Podocyte-specific deficiency of MAGI-2 in mice abrogated localization of Nephrin and Neph1 independently of other scaffold proteins"* (PMID: 33144214), and that MAGI-2 downregulation coincides with reduced slit-diaphragm backbone proteins in human FSGS and IgA nephropathy. Ihara et al., 2014 (*Am J Pathol*) showed that *"homozygous deletion of the Magi2 gene in mice caused neonatal lethality, which was explained by podocyte morphological abnormalities and anuria"* (PMID: 25108225), with decreased nephrin and dendrin at the slit diaphragm, nuclear translocation of dendrin, and enhanced cathepsin L expression. These establish MAGI2's role in positioning the slit-diaphragm backbone and the severity gradient of complete versus partial loss.

Finding 8 — Strong loss-of-function constraint; ultra-rare recessive disorder (F008)

gnomAD constraint metrics for MAGI2 (ENSG00000187391; GRCh38 chr7:78,017,055–79,453,667; 7q21.11) indicate strong depletion of heterozygous loss-of-function alleles: **pLI** = **1.00**, **LOEUF** = **0.35**, observed/expected pLoF = 35/133.3 (oe = 0.26), LoF Z = 7.22. Despite this constraint — which indicates heterozygous LoF is under strong purifying selection — NS15 manifests only with **biallelic** LoF. Only a handful of families have been reported (3 patients/2 families in the index report; additional families in the pTSNS cohort), making NS15 ultra-rare with no established prevalence or incidence figures.

Finding 9 — MAGI2 links slit diaphragm to actin and protects against apoptosis; AAV-MAGI2 is protective (F009)

Asanuma-group work (2026) shows that *"MAGI-2 binds to Synpo and α-actinin-4, regulating actin filament localization and contributing to cytoskeletal reorganization in podocytes"* (PMID: 41182836); podocyte-specific MAGI-2 loss mislocalizes actin bundles from the foot-process center toward the basement membrane, driving foot-process effacement. In diabetic kidney disease models, MAGI2 is downregulated by high glucose, activating TGF-β-Smad3 signaling and reducing anti-apoptotic proteins; critically, *"enforced expression of MAGI2 via AAV*

transduction protected podocytes from apoptosis, with concomitant improvement of renal functions" (PMID: 37950637). MAGI2 is also a candidate biopsy biomarker downregulated in injured podocytes, discriminating primary FSGS from minimal change disease (PMID: 40563084).

Finding 10 — Identifiers and MAGUK scaffold architecture (F010)

MAGI2 = *membrane-associated guanylate kinase, WW and PDZ domain containing 2*. Identifiers: **HGNC:18957; NCBI Gene 9863; Ensembl ENSG00000187391; UniProt Q86UL8** (MAGI2_HUMAN, 1455 aa); gene OMIM 606382; cytogenetic location 7q21.11 (GRCh38 chr7:78,017,055–79,453,667). Aliases: AIP1, ARIP1, ACVRIP1, KIAA0705, MAGI-2. Protein domain architecture (UniProt): an **inverted MAGUK** with N-terminal PDZ0 (aa 17–101), a guanylate kinase-like domain (109–283), two WW domains (302–335; 348–381), and six PDZ domains (PDZ1–6). Orthologs: mouse *Magi2* (NCBI 50791), rat *Magi2* (113970), zebrafish *magi2a* and *magi2b*. Disease (phenotype) OMIM: **617609** (Nephrotic syndrome, type 15).

Report by Section

1. Disease Information

NS15 is a Mendelian form of **steroid-resistant nephrotic syndrome** — a kidney disorder of the glomerular filtration barrier characterized by heavy proteinuria, hypoalbuminemia, edema, and hyperlipidemia that fails to remit with corticosteroid therapy. It belongs to the broad family of genetic podocytopathies/FSGS.

Resource	Identifier
MONDO	MONDO:0033262
OMIM (phenotype)	617609 (Nephrotic syndrome, type 15)
OMIM (gene)	606382 (MAGI2)
HGNC	HGNC:18957
NCBI Gene	9863
Ensembl	ENSG00000187391
UniProt	Q86UL8
MeSH (parent)	Nephrotic Syndrome (D009404); Nephrosis, Lipoid / FSGS
ICD-10	N04 (Nephrotic syndrome)
ICD-11	GB41 (Nephrotic syndrome)

Synonyms: nephrotic syndrome, type 15; NPHS15; MAGI2-associated nephrotic syndrome/SRNS; MAGI2-related congenital nephrotic syndrome. Information is derived from **aggregated disease-level resources** (OMIM, individual case reports, and small family cohorts) rather than EHR; the total reported patient population is on the order of a few dozen.

2. Etiology

Primary cause — genetic. NS15 is caused exclusively by **biallelic loss-of-function variants in *MAGI2*** (autosomal recessive). There is no infectious or primary environmental etiology for the Mendelian disease.

Genetic risk factors. Causal variants are rare biallelic truncating/LoF alleles (frameshift, nonsense, read-through, duplication). Being a **carrier** (heterozygous) confers no overt kidney disease (recessive). Consanguinity is a recognized contributor given homozygous frameshift variants in sibling cases (PMID: 27932480). MAGI2 sits within a functional network (TNS2, DLC1, CDK20, ITSN1, ITSN2) where other genes cause related nephrotic phenotypes (PMID: 29773874).

Environmental risk / protective factors. Not established for the Mendelian form. In acquired podocytopathies, MAGI2 downregulation is induced by hyperglycemia (TGF-β/Smad3 activation) (PMID: 37950637) — relevant to gene–environment interplay in common podocyte injury but not to the monogenic disease per se.

Gene–environment interactions. In acquired injury, corticosteroids paradoxically *exacerbate* the phenotype in zebrafish *magi2a* knockouts (PMID: 31010479), and dexamethasone modulates RhoA within the MAGI2 network (PMID: 29773874) — a GxE signal with therapeutic relevance.

3. Phenotypes

Phenotype	Type	HPO term	Onset	Severity	Frequency
Proteinuria / nephrotic-range proteinuria	Lab abnormality	HP:0000093 (Proteinuria); HP:0003076 (Glomerular proteinuria)	Congenital–childhood	Severe	Universal (defining)
Steroid-resistant nephrotic syndrome	Clinical sign	HP:0000100 (Nephrotic syndrome); HP:0004722 (steroid-resistant)	Congenital–childhood	Severe	Universal
Hypoalbuminemia	Lab abnormality	HP:0003073	Early	Moderate–severe	High
Edema	Clinical sign	HP:0000969	Early	Variable	High
Focal segmental glomerulosclerosis	Histopathology	HP:0000097	Variable	Progressive	Common
Diffuse mesangial sclerosis	Histopathology	HP:0001967	Congenital	Severe	Reported
Progression to ESKD / renal failure	Clinical outcome	HP:0003774 (Stage 5 CKD); HP:0000083 (Renal insufficiency)	Variable	Severe	Frequent
Hyperlipidemia	Lab abnormality	HP:0003077	Secondary	Moderate	Common

Progression is typically progressive toward kidney failure. **Quality-of-life impact** is substantial: nephrotic syndrome in children imposes edema, infection risk, hospitalization, dialysis dependence, and transplant — major effects on growth, schooling, and family function. No MAGI2-specific EQ-5D/SF-36 data exist.

4. Genetic / Molecular Information

Causal gene: *MAGI2* (HGNC:18957; gene OMIM 606382). **Variant classification (ACMG/AMP):** reported variants are **pathogenic/likely pathogenic** truncating alleles. **Variant types:** frameshift, nonsense (premature stop), stop-loss/read-through, and duplication (PMID: 27932480). **Zygosity:** homozygous or compound heterozygous (biallelic). **Functional consequence: loss of function** — loss of or diminished podocyte MAGI2 protein by immunohistochemistry; patient variants additionally lose RapGEF2 binding and Rap1 activation (PMID: 31171376).

Allele frequency: causal alleles are extremely rare/absent in gnomAD; the gene is strongly LoF-constrained (pLI = 1.00, LOEUF = 0.35) (F008). **Origin:** germline. **Modifier genes:** the physically interacting network (TNS2, DLC1, CDK20, ITSN1, ITSN2) may modify expression (PMID: 29773874); **dendrin** dosage modifies severity — dendrin ablation prolongs lifespan of MAGI2-podKO mice (PMID: 37134307). **Epigenetic / chromosomal abnormalities:** none specific to NS15 reported; MAGI2 is a WT1 and MafB transcriptional target (PMID: 25556170; PMID: 32622525).

5. Environmental Information

No environmental toxin, radiation, lifestyle, or infectious agent causes the **Mendelian** disease. Relevant to acquired MAGI2 loss: **hyperglycemia** downregulates MAGI2 via TGF-β/Smad3 (PMID: 37950637). Corticosteroids worsen MAGI2-deficient podocytes in zebrafish (PMID: 31010479). No infectious agents implicated.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic MAGI2 LoF variants** (frameshift/nonsense/read-through) **result in** absent or non-functional MAGI2 scaffold protein in podocytes (demonstrated; IHC). (PMID: 27932480)
2. Loss of MAGI2 scaffold **leads to** failure to localize/anchor slit-diaphragm backbone proteins **nephrin (NPHS1) and Neph1** at the junction (demonstrated in mouse). (PMID: 33144214, PMID: 25271328)
3. In parallel, loss of MAGI2 **disrupts** the MAGI2–RapGEF2 complex, **resulting in** failure to activate the small GTPase **Rap1** (demonstrated; patient variants lose RapGEF2 binding). (PMID: 31171376)
4. **Branch A (cytoskeleton):** collapsed Rap1 and **Rho-like GTPase (RhoA/Rac1/Cdc42)** signaling, plus loss of MAGI2 binding to **synaptopodin and α-actinin-4**, **lead to** actin

cytoskeletal disorganization and mislocalization of actin bundles toward the basement membrane. (PMID: 31171376, PMID: 29773874, PMID: 41182836)

5. **Branch B (apoptosis):** MAGI2 loss **decreases dendrin** at the slit diaphragm and **promotes nuclear translocation of dendrin**, which **leads to** JNK phosphorylation, altered focal adhesion, and detachment-induced podocyte **apoptosis**; TGF-β/Smad3 activation **reduces anti-apoptotic proteins** (demonstrated in mouse/in vitro). (PMID: 28539383, PMID: 37134307, PMID: 37950637)
6. Branches A and B converge: cytoskeletal collapse and apoptosis **result in foot-process effacement and podocyte loss/depletion**. (PMID: 41182836)
7. Podocyte depletion **leads to** breakdown of the filtration barrier → **massive proteinuria** (nephrotic range). (PMID: 28539383)
8. Sustained podocyte loss **results in glomerulosclerosis (FSGS / diffuse mesangial sclerosis)**, rising creatinine, and progression to **ESKD/renal failure** (demonstrated; mouse time course and human). (PMID: 28539383)

```
MAGI2 biallelic LoF
        |
   loss of podocyte MAGI2 scaffold
        |
   +----+--------------------------+
   |                               |
 nephrin/Neph1             MAGI2–RapGEF2–Rap1
 mislocalization           + RhoA/Rac1/Cdc42 collapse
   |                        + synaptopodin/alpha-actinin-4 unbinding
   |                               |
   |                      actin cytoskeleton disorganization (Branch A)
   |                               |
   |   dendrin down + nuclear dendrin -> JNK -> apoptosis (Branch B)
   |   TGF-beta/Smad3 up -> anti-apoptotic proteins down
   +----------------+-------------+
                    |
         foot-process effacement + podocyte loss
                    |
             nephrotic-range proteinuria
                    |
         glomerulosclerosis (FSGS/DMS) -> ESKD
```

Pathways/processes & ontology terms. Molecular pathways: Rap1 signaling (KEGG hsa04015), Rho GTPase cycle (Reactome), TGF-β/Smad3, slit-diaphragm phase separation (PMID: 34330769). GO biological processes: **GO:0036057** (slit diaphragm assembly), **GO:0030036** (actin cytoskeleton organization), **GO:0007265** (Ras/small-GTPase signal transduction),

GO:0006915 (apoptotic process), **GO:0072015** (glomerular visceral epithelial cell development). GO cellular components: **GO:0036056** (filtration diaphragm), **GO:0015629** (actin cytoskeleton), **GO:0005886** (plasma membrane). Cell type: **CL:0000653** (podocyte / glomerular visceral epithelial cell).

7. Anatomical Structures Affected

- **Primary organ:** kidney (UBERON:0002113); specifically the **renal glomerulus** (UBERON:0000074) and the **glomerular filtration barrier**.
- **Cell type: podocyte** (CL:0000653); subcellular focus on **foot processes** and the **slit diaphragm** (specialized intercellular junction; GO:0036056).
- **Subcellular compartments:** plasma membrane junction (slit diaphragm), actin cytoskeleton, nucleus (dendrin translocation).
- **Secondary involvement/body systems:** urinary/renal system; secondary cardiovascular (edema, hyperlipidemia), hematologic (hypercoagulability of nephrotic syndrome), and immune (infection susceptibility) consequences.
- **Lateralization: bilateral** (both kidneys), diffuse glomerular involvement.

8. Temporal Development

Onset: congenital to childhood; complete loss is neonatally lethal in mice (PMID: 25108225), and human cases present as congenital/early-childhood SRNS. **Onset pattern:** insidious/chronic proteinuria that may present acutely with edema. **Progression:** progressive; mouse models show albuminuria → glomerulosclerosis → rising creatinine → death across weeks (8→12→20 wk in podocyte-specific KO) (PMID: 28539383). **Course:** chronic, lifelong, progressing to ESKD. **Remission:** does not remit spontaneously or reliably with immunosuppression; kidney transplant resolves the filtration defect without recurrence (PMID: 32317330). **Critical period:** podocyte development and maintenance; MAGI2 is a WT1/MafB target required for podocyte differentiation (PMID: 25556170).

9. Inheritance and Population

Inheritance: autosomal recessive; biallelic LoF required. **Penetrance:** appears complete for biallelic LoF (all reported biallelic cases affected). **Expressivity:** variable (congenital DMS to childhood FSGS; some families "partially treatment-sensitive," PMID: 29773874). **Carrier status:** heterozygotes unaffected; carrier frequency not formally established but LoF alleles are extremely rare (gnomAD pLI = 1.00, F008). **Consanguinity:** contributory (homozygous

frameshift in siblings, PMID: 27932480). **Founder effects / anticipation / mosaicism:** none reported. **Epidemiology:** ultra-rare — only a handful of families; no reliable prevalence/incidence. **Sex ratio:** no sex bias expected (autosomal recessive). **Geographic distribution:** no established pattern; cases are sporadic across populations.

10. Diagnostics

Clinical/laboratory: nephrotic-range proteinuria (urine protein:creatinine), hypoalbuminemia, hyperlipidemia; elevated creatinine with progression. **Biopsy:** FSGS or diffuse mesangial sclerosis on light microscopy; **foot-process effacement** on electron microscopy; reduced/absent MAGI2 and reduced slit-diaphragm backbone proteins by immunohistochemistry (PMID: 33144214). MAGI2 is itself a candidate biopsy biomarker discriminating primary FSGS from minimal change disease (PMID: 40563084).

Genetic testing — the definitive diagnostic. Recommended approach: gene-panel or exome/genome sequencing in all pediatric SRNS. Cohort data: monogenic cause in ~30% of childhood SRNS (PMID: 32155690); WGS identified monogenic disease in 56.1% of early-onset SRNS families (PMID: 40402239); WES adds ~8.7% over panels, WGS adds ~4.3% over WES, but panels are most cost-effective first-line (PMID: 38654395). *MAGI2* is included on SRNS/FSGS NGS panels. **Differential diagnosis:** other monogenic SRNS genes (NPHS1, NPHS2, WT1, PLCE1, LAMB2, TRPC6, etc.) and acquired FSGS/minimal change disease.

11. Outcome / Prognosis

Prognosis is poor for native kidneys — progression to ESKD is typical. In the mouse analog, 100% mortality by 20 weeks (PMID: 28539383). In humans, monogenic SRNS responds poorly to immunosuppression (~4% complete response) (PMID: 32317330). **The key favorable prognostic feature is absence of post-transplant recurrence**, so kidney transplantation offers durable renal replacement without recurrent disease (PMID: 32317330). **Complications:** nephrotic-syndrome sequelae (thromboembolism, infection, hyperlipidemia, growth failure) and ESKD-related morbidity. **Prognostic factors:** monogenic status itself is the dominant prognostic marker; degree of residual MAGI2 function may modulate treatment sensitivity.

12. Treatment

No disease-specific approved therapy exists. Management mirrors genetic SRNS:

Modality	Detail	NCIT suggestion
Corticosteroids	Generally ineffective (steroid-resistant); may worsen MAGI2-deficient podocytes in models	NCIT:C305 (Corticosteroid)
Calcineurin inhibitors (tacrolimus)	Partial response in some monogenic/pTSNS; steroids+TAC most effective overall in SRNS cohorts (PMID: 38654395)	NCIT:C1000 (Tacrolimus)
RAAS blockade (ACEi/ARB)	Antiproteinuric/renoprotective supportive care	NCIT:C77537 (ACE inhibitor)
Supportive care	Diuretics, albumin, anticoagulation, lipid control, infection prophylaxis	NCIT:C15277 (Supportive Care)
Kidney transplantation	Definitive renal replacement; **no recurrence** in monogenic SRNS (PMID: 32317330)	NCIT:C15366 (Kidney Transplantation)
Experimental: Rap1 activation	Full rescue of actin defect in patient podocytes (PMID: 31171376)	—
Experimental: AAV-MAGI2 gene replacement	Protected podocytes from apoptosis, improved renal function in models (PMID: 37950637)	NCIT:C15288 (Gene Therapy)
Experimental: importin-α inhibition (ivermectin)	Blocks dendrin nuclear translocation, reduces podocyte loss/glomerulosclerosis in MAGI2-podKO mice (PMID: 37134307)	—

Pharmacogenomics: the monogenic diagnosis predicts non-response to immunosuppression and favorable transplant outcome, guiding avoidance of unnecessary immunosuppressive toxicity (PMID: 32317330).

13. Prevention

No primary prevention exists (genetic disease). **Secondary/tertiary:** early genetic diagnosis to avoid futile/harmful immunosuppression, optimize renoprotection, and plan transplantation. **Genetic counseling:** autosomal recessive — 25% recurrence risk per pregnancy for carrier couples; cascade carrier testing and, where desired, prenatal/preimplantation genetic testing are applicable. **Screening:** targeted sequencing of *MAGI2* in SRNS; no population newborn screening.

14. Other Species / Natural Disease

Orthologs/models: mouse *Magi2* (NCBI 50791; NCBI Taxon 10090), rat *Magi2* (113970; Taxon 10116), zebrafish *magi2a*/*magi2b* (Taxon 7955). Mouse and zebrafish knockouts recapitulate proteinuria/glomerulosclerosis (PMID: 28539383; PMID: 31010479). No naturally occurring companion-animal or wildlife MAGI2 nephropathy is documented in OMIA. Mechanisms are evolutionarily conserved (zebrafish podocyte requirement). No zoonotic potential (non-infectious).

15. Model Organisms

Model	Type	Phenotype recapitulation	Reference
Podocyte-specific *Magi2* KO mouse	Conditional KO (mammalian)	Slit-diaphragm disruption, albuminuria (8 wk), glomerulosclerosis (12 wk), death (20 wk) — excellent	PMID: 28539383
Constitutive *Magi2*-null mouse	Global KO	Proteinuria by 2 wk, nephrin loss, renal failure death by ~3 mo	PMID: 25271328
Homozygous *Magi2* deletion mouse	Global KO	Neonatal lethality, podocyte abnormalities, anuria, dendrin/ cathepsin L changes	PMID: 25108225
magi2a / *magi2b* zebrafish	CRISPR KO	*magi2a* KO → proteinuria/nephrotic phenotype; steroids exacerbate	PMID: 31010479
Cultured human podocytes + patient variants	In vitro	Actin disorganization, rescued by Rap1 activation	PMID: 31171376

Applications: slit-diaphragm assembly, Rap1/RhoA signaling, dendrin apoptosis, drug screening, gene therapy. **Limitations:** complete-KO neonatal lethality limits adult study (requires conditional/podocyte-specific alleles); mouse does not capture full human allelic heterogeneity. Resources: MGI, ZFIN, IMPC.

Mechanistic Model / Interpretation

MAGI2 is best understood as the **organizing scaffold of the podocyte slit diaphragm**, performing three non-redundant jobs whose simultaneous failure explains the disease: (1) **positioning** the junctional backbone (nephrin/Neph1) and nucleating the slit-diaphragm condensate via liquid–liquid phase separation with dendrin/CD2AP (PMID: 34330769); (2) **signaling** through RapGEF2–Rap1 and the RhoA/Rac1/Cdc42 network to maintain the actin cytoskeleton (PMID: 31171376, PMID: 29773874); and (3) **survival** by restraining dendrin nuclear translocation and TGF-β/Smad3-driven apoptosis (PMID: 37134307, PMID: 37950637). Upstream lies the mutation; the earliest downstream nodes are scaffold loss and Rap1 collapse; the terminal events are podocyte depletion and sclerosis. The full rescue of patient-podocyte phenotypes by Rap1 activation and the protection by AAV-MAGI2 establish that the pathway is reversible at the cellular level — the strongest argument for future targeted therapy.

Evidence Base

PMID	Contribution	Relationship to findings
31171376	MAGI2–RapGEF2–Rap1 axis; patient podocyte rescue	Supports F001, F002 (core mechanism)
28539383	Podocyte-specific KO time course; dendrin	Supports F003
25271328	Constitutive KO; nephrin loss	Supports F003, F007
25108225	Neonatal lethality; dendrin/cathepsin L	Supports F007
33144214	Nephrin/Neph1 localization	Supports F007
27932480	Index human variant report	Supports F005
29773874	RhoA GTPase network; pTSNS	Supports F006
32317330	Immunosuppression response; no transplant recurrence	Supports F004
41182836	Synaptopodin/α-actinin-4 binding	Supports F009
37950637	TGF-β/Smad3 apoptosis; AAV-MAGI2	Supports F009
37134307	Dendrin nuclear translocation; importin-α inhibition	Mechanism & experimental therapy
34330769	Slit-diaphragm phase separation	Mechanism (condensate)
31010479	Zebrafish magi2a KO; steroid exacerbation	Model & GxE
25556170	MAGI2 as WT1/MafB target	Transcriptional regulation

Limitations and Knowledge Gaps

- **Very small human sample.** Causality rests on ~3 patients/2 families in the index report plus additional pTSNS families; no prevalence/incidence, penetrance estimates, or natural-history cohorts exist.

- **Genotype–phenotype uncertainty.** The spectrum from congenital diffuse mesangial sclerosis to "partially treatment-sensitive" childhood NS is not fully explained; modifier contributions (dendrin, network genes) are inferred.
- **Mechanism largely from models.** Branch-level steps (dendrin apoptosis, TGF-β/Smad3, actin mislocalization) are demonstrated primarily in mouse/zebrafish/in vitro and in **acquired** injury contexts (diabetic kidney disease), not directly in NS15 patient tissue.
- **No human therapeutic data.** Rap1 activation, AAV-MAGI2, and importin-α inhibition are preclinical only.
- **Citation verification.** Several supporting snippets (F005–F007, F009) were drawn from abstract text that should be re-verified against source abstracts before database ingestion.

Proposed Follow-up Experiments / Actions

1. **GeneMatcher/registry aggregation** of additional MAGI2 biallelic families to quantify penetrance, onset distribution, and genotype–phenotype correlation.
2. **Patient-derived podocytes/organoids/iPSC** carrying truncating variants to test Rap1 activators and importin-α inhibitors (ivermectin) as repurposed therapies.
3. **AAV-MAGI2 dose-finding** in conditional-KO mice to define a therapeutic window before irreversible sclerosis.
4. **Biopsy MAGI2 immunostaining** validation as a prognostic/diagnostic biomarker across SRNS subtypes.
5. **Verification** of the unchecked citation snippets against source abstracts prior to knowledge-base entry.
